

Machine Learning Tasks, Data Architecture, Learning Problems, and Evaluation

Prepared for Mechanical Engineering students beginning machine learning study

This reading introduces two linked topics. Subject 1 explains the main kinds of machine learning tasks and shows how mechanical engineering data is organized for learning. Subject 2 explains common learning problem types, practical evaluation metrics, and the data pitfalls that can quietly damage a project. The focus and examples are framed for real mechanical engineering work.

Learning goals: By the end of this reading, you should be able to identify the main ML task type in a mechanical engineering problem, describe one datapoint in tabular, image, or time-series form, explain the roles of training, validation, and test data, distinguish supervised, unsupervised, and semi-supervised learning, and choose evaluation metrics that match safety, cost, or reliability goals.

Reference basis. This tutorial follows standard machine learning terminology and evaluation guidance from the [scikit-learn](#) User Guide and Google's Machine Learning Crash Course, while the mechanical engineering examples are aligned with recent review literature on AI and ML applications in mechanical engineering [1]-[10].

Acronyms & Parameters

ML = Machine Learning	R^2 = R-squared	Y = output variables
AI = Artificial Intelligence	F1 = F1 score	mm = millimeters
CCTV = Closed-Circuit Television	TP = True Positives	MPa = megapascals
MAE = Mean Absolute Error	FP = False Positives	m^3/s = cubic meters per second
MSE = Mean Squared Error	FN = False Negatives	
RMSE = Root Mean Squared Error	X = input features / inputs	
	y = output / target / label	

Subject 1. ML Tasks and Data Architecture

Machine learning is most useful when mechanical engineers have repeated examples, measurable variables, and a clear decision, forecast, or pattern to automate. At a high level, many applications can be grouped into four task families: prediction, recognition, feature extraction, and acceleration.

Table 1. Core ML task types with mechanical engineering examples

Task family	What the model does	Mechanical engineering example	Typical output
Prediction / estimation	Predict a number or category from measured inputs	Estimate mechanical material strength from mix variables; predict cooling-system overload depth from flow-rate and terrain; forecast production flow	One value or one class
Recognition	Identify whether a pattern is present and often where it is located	Detect cracks or surface defects in images; identify damaged mechanical components; flag pipe defects from CCTV frames	Class, box, or mask
Feature extraction	Turn raw measurements into a smaller and more informative representation	Extract damage-sensitive indicators from vibration signals; compress many sensor channels into a few state variables	New features
Acceleration / surrogate modeling	Replace a slow repeated computation with a faster approximation	Approximate many finite element runs; accelerate cooling-system overload inundation maps; emulate repeated design studies	Fast estimate of a simulation or response

The four task families often overlap. For example, a crack inspection system may first perform recognition on images, then extract features from the detected crack pattern, and finally predict deterioration severity or maintenance priority. Recent mechanical engineering reviews show broad use of these task types across structures, mechanics of materials, vehicle systems, thermal-fluid systems, and manufacturing systems management [10].

1.1 What is one datapoint?

A datapoint is one example given to the model. In **supervised learning**, one datapoint usually contains inputs X and output y. Inputs are also called features, variables, or attributes. The output is also called the target or label. Some problems have one output, while others have several. Figure 1 shows the basic idea for a tabular repository.

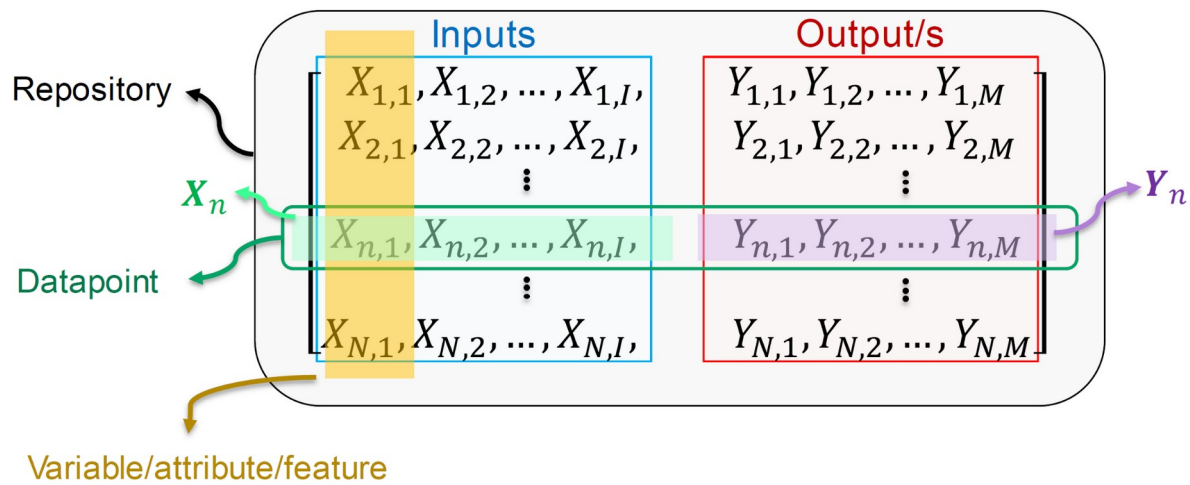


Figure 1. Conceptual view of a repository, one datapoint, its input features X , and its output variables Y .

For a material strength model, one datapoint could be one mix design and its tested strength. For a mechanical component image classifier, one datapoint could be one image and its crack label. For a production throughput or vibration model, one datapoint could be one time window and the response linked to that window. The key idea is always the same: define one consistent unit of observation before any modeling starts. Note that having inputs is mandatory, but outputs are not, and it depends on the task at hand and the learning problem.

1.2 Common data formats in mechanical engineering

Mechanical engineering ML projects most often start from tabular data, image data, or time-series data. Each format can represent the same physical system in a different way, so the format should match the engineering question. Figure 2 summarizes the three common formats.

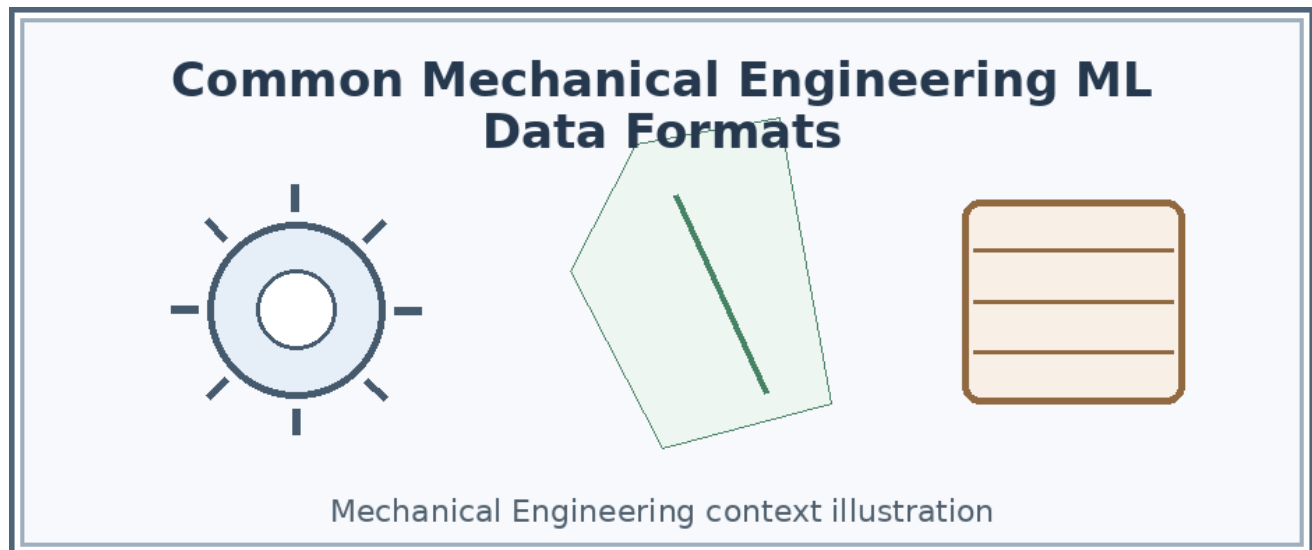


Figure 2. Examples of one datapoint in tabular, image, and time-series form for mechanical engineering problems.

Table 2. How one datapoint changes across common data formats

Format	What one datapoint looks	Mechanical engineering example	When it fits well
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	like		
Tabular	Rows are samples; columns are features	Material mix variables, mechanical component inventory tables, machined-surface defect summaries	Best when each case can be summarized by measured variables
Image	Pixels form the input array	Drone images of cracks, wind tunnel test section lining photos, and machined surface images	Best when spatial appearance matters
Time-series	Ordered sequence over time	Acceleration records, strain histories, flow-rate-flow series, and production throughput speed	Best when timing and sequence matter

Practical tip: Before building any model, write one sentence that defines the datapoint. For example: 'One datapoint is one turbine blade surface image,' or 'One datapoint is one 10-minute sensor window.' This simple step prevents many later mistakes. Prediction problems are the easiest place to start because the question is clear: given inputs, what output should we estimate? Recognition and feature extraction are common when the data are images or sensor signals. Acceleration becomes important when a physical simulation is accurate but too slow to run many times.

1.3 Training, validation, and test data

Training data is used to fit model parameters. Validation data is used during development to compare settings, features, or model choices. Test data is kept aside for the final check on unseen examples [4,8]. Testing on the same data used for training can make a weak model look excellent, which is why a clean split is essential [4]. Think of an ML model as a student: she studies the training materials and then is tested on the exam with test data. This student has parameters, constraints, and limitations. Learning is about gradually finding the optimal parameters so that the task is performed well, becoming comfortable with the training materials, and achieving a good score on the exam using the test data.

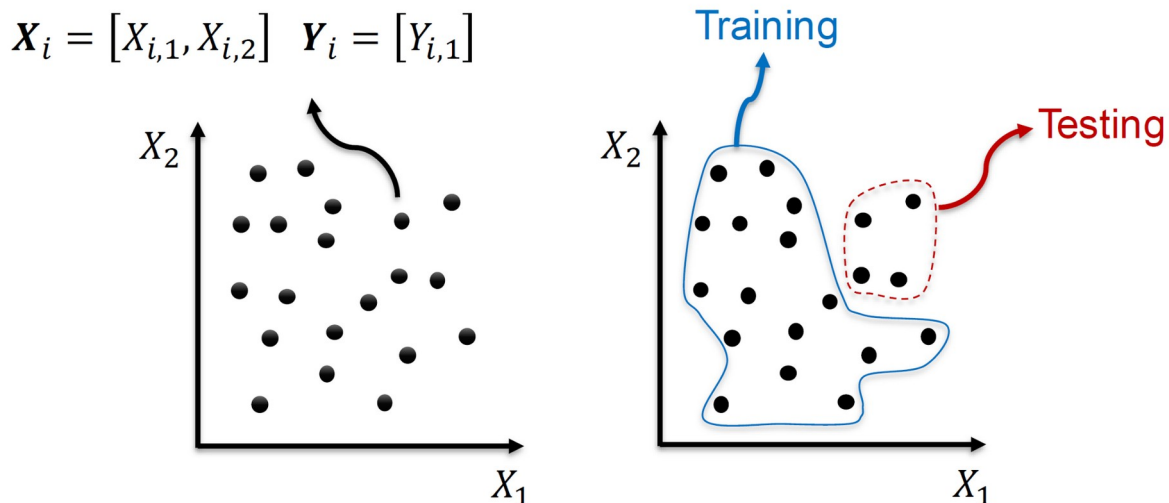


Figure 3. One example of separating datapoints into training and testing subsets in the same feature space.

A three-way split is often more useful than a simple train-test split because it keeps the test set for the final report and uses the validation set for repeated tuning [8]. Validation is like our student setting aside a part of the training materials and using them to keep validating himself throughout the semester, leading up to exam day. There is no single mandatory percentage, but training is usually the largest subset. The key requirement is that validation and test examples should represent the kind of data the model will face later [8].

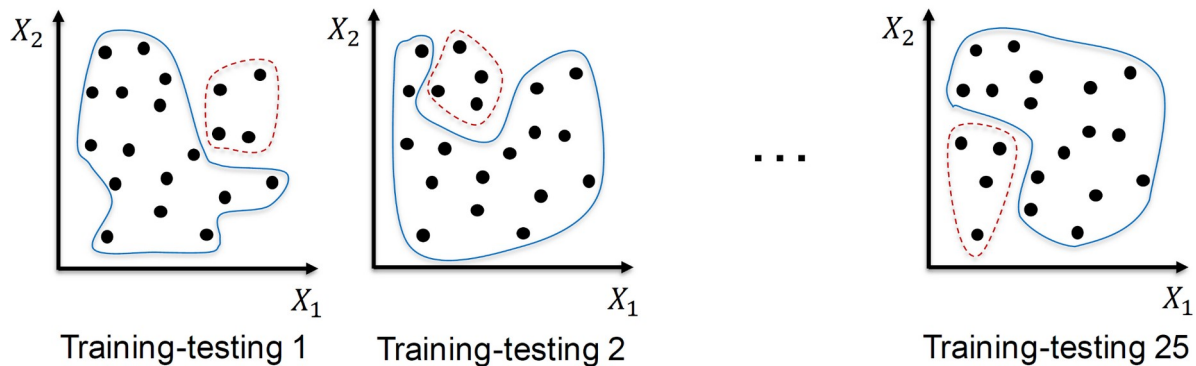


Figure 4. Repeated random train-test divisions illustrate why one lucky split should not be trusted by itself.

Figure 4 also shows why a single score from a single split is not sufficient when the data is limited. Repeated random splits or [cross-validation](#) help estimate whether the model is consistently strong or only benefited from an easy split [4]. Imagine if our student (ML model) is given easy training materials and encounters a hard test on exam day, or hard training data and an easy test on exam day; in both cases, it's unfair. Instead, imagine students are repeatedly trained and examined, each time using a different train-test set; the average student's training and testing performance would be a good representation of the student's capability. This matters in mechanical engineering because datasets are often small, expensive to label, and collected under narrow field conditions.

Mechanical engineering splitting rule: Split by the real deployment unit whenever possible. If the model will be used on new mechanical components, split by mechanical component. If it will be used on future storms, split by event or by time. If records from the same mechanical component, site, or day appear in both training and test sets, the evaluation can look better than true field performance.

Subject 2. Learning Problems and Evaluation

The first question in any ML project is not 'Which algorithm should I use?' but rather 'What kind of learning problem do I have?' Problem type determines the data you need, the output you expect, and the metric that should guide decisions.

2.1 Supervised, unsupervised, and semi-supervised learning

Supervised learning uses labeled outputs. scikit-learn groups regression and classification under supervised learning [1]. Unsupervised learning works with inputs only and looks for structure such as clusters or lower-dimensional representations [2]. Semi-supervised learning sits between them: only some training examples are labeled, which is useful when labeling is expensive but raw data is plentiful [3].

Table 3. Common learning problem types and their evaluation style

Problem type	Data structure	Mechanical engineering example	Typical evaluation
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Supervised regression	Inputs X and numeric output y	Predict mechanical component deflection, cooling-system overload depth, compressive strength, deflection	MAE, MSE, RMSE, R^2
Supervised classification	Inputs X and categorical y	Crack / no-crack, safe / warning / danger, material class	Accuracy, precision, recall, F1, confusion matrix
Unsupervised clustering	Inputs only	Group similar production throughput patterns, vibration states, or zones with similar mechanics of materials and structures behavior	Silhouette score, stability, engineering usefulness
Unsupervised dimensionality reduction	Inputs only	Compress many sensor channels into a few indicators or visual axes	Explained variance, reconstruction error, downstream usefulness
Generative problems	Usually inputs only, sometimes conditional labels	Generate realistic crack images, synthetic sensor traces, or design candidates	Sample quality, reconstruction error, expert review, downstream value
Semi-supervised learning	Small labeled set plus large unlabeled set	Few labeled defect images and many unlabeled inspection images	Use the metric of the final supervised task

2.2 Regression and classification

Regression predicts a number. Examples include predicting compressive strength, travel time, wear depth, bearing capacity, or pipe flow. Classification predicts a category. Examples include whether a mechanical component is damaged, which machined-surface defect type is present, or whether a slope is low, medium, or high risk. These are the two most common starting points for beginner mechanical engineering projects [1,10].

A useful habit is to write the output in plain language before selecting the model. If the output sentence starts with 'how much' or 'how many,' regression is likely. If it starts with 'which class' or 'which category,' classification is likely.

2.3 Unsupervised and semi-supervised learning

Unsupervised learning is valuable when no agreed-upon labels are available yet. Clustering can reveal groups of mechanical components, production throughput states, or material behaviors that look similar. Dimensionality reduction can compress many measured variables into a smaller set that is easier to visualize or model [2]. Generative models go one step further and try to learn how the data itself is distributed so they can produce realistic new samples or reconstructions.

Semi-supervised learning is often practical in mechanical engineering because expert labels are expensive. For example, an engineer may have labels for only a small number of defect images or corrosion states, while thousands of unlabeled images are available. Semi-supervised methods try to use both types of data together [3].

Beginner rule: If you have reliable labels and a clear target, start with supervised learning. If labels are scarce or unclear, use unsupervised exploration first and treat the results as engineering clues, not final truth.

2.4 Core evaluation metrics

Table 4. Beginner-friendly metric cheat sheet

Metric	Simple formula or idea	What it tells you	When it is especially useful
MAE	Average absolute error	Average miss measured in the original unit	Easy to interpret when units matter, such as mm, MPa, or m ³ /s
MSE	Average squared error	Penalizes large mistakes strongly	Useful when large misses are especially costly
RMSE	Square root of MSE	Large errors still matter more, but the result returns to the original unit	Common for prediction reports
Accuracy	Correct predictions / all predictions	Overall fraction correct	Only reliable when classes are reasonably balanced
Precision	TP / (TP + FP)	Of all positive alerts, how many were truly positive?	Best when false alarms are costly
Recall	TP / (TP + FN)	Of all true positives, how many were found?	Best when missed detections are dangerous
F1 score	2PR / (P + R)	Balances precision and recall	Useful for imbalanced classification when both types of error matter
Silhouette score	Cluster separation measure	Higher values mean points fit their own cluster better than nearby clusters	Useful for clustering, but still needs engineering interpretation

Metric choice should follow the final engineering decision, not just textbook habit. scikit-learn's evaluation guide emphasizes starting from the application goal and the decision that will be made from the prediction [5]. Google's ML Crash Course similarly notes that accuracy alone is a poor guide on imbalanced problems, and that precision, recall, and F1 often give better insight [7].

2.5 Choosing metrics for safety, cost, and reliability

Table 5. Match the metric to the engineering consequence

Application	Main concern	Metrics to prioritize	Why
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Mechanical component damage screening	Missing real damage	Recall, then F1	A false negative can delay inspection of a true defect
Automatic defect alerts for maintenance crews	Too many false alarms	Precision	Low precision can waste field time and budget
Cooling-system overload depth prediction	Large numeric misses	MAE and RMSE	Report both average miss and sensitivity to large misses
Production-state classification	Confusing rare bottleneck states	Per-class recall and confusion matrix	Overall accuracy can hide poor performance on rare but important classes
Material strength prediction	Outliers that are unsafe for design	MAE, RMSE, error plots near limits	Look beyond one average score
Condition rating models for assets	Uneven performance across asset groups	Per-group metrics and calibration checks	A fair average may still hide weak performance on one subgroup

In short, recall matters when missing a positive case is dangerous, precision matters when false alarms are expensive, and F1 is a useful compromise when both matter [7]. For regression, MAE is often the easiest to explain, while MSE or RMSE become more important when large misses are unacceptable. For unsupervised tasks, no single number should replace engineering judgment.

2.6 Data pitfalls in mechanical engineering ML

Table 6. Common data pitfalls and how to spot them

Pitfall	What it means	Mechanical engineering example	Basic safeguard
Sampling bias	Training data comes from a narrow or unrepresentative sample	Only urban mechanical components, only daytime production throughput, only dry-season imagery	Collect broader data or state limits clearly [9]
Distribution shift	Deployment data differs from development data	New camera angle, different sensor brand, extreme weather year, new site geology	Test on truly future or out-of-site data [8]
Data leakage	Information from the future or the test set sneaks into training	Using the same mechanical component in both sets, using future measurements to predict past failure	Split by asset, site, or time before feature engineering [4,8]
Inconsistent preprocessing	Transformations are not applied the same way to all sets	Scaling training data but forgetting to scale test data	Use one repeatable preprocessing pipeline [6]
Overfitting	Model memorizes training details and fails to generalize	Excellent training score but unstable field performance	Use validation, cross-validation, simpler baselines, and error inspection [4]

Sampling bias occurs when the collected data is not representative [9]. In mechanical engineering, that might mean a mechanical component dataset dominated by one climate region or one mechanical component type. Distribution shift occurs when the model is deployed on data that no longer looks like the development data, such as after a sensor replacement, a change in lighting, or an extreme weather event [8]. These two issues are especially important in safety-critical and site-specific applications [10].

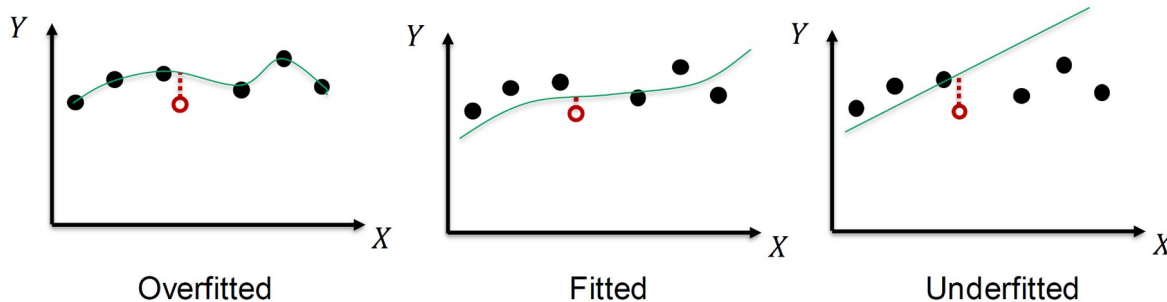


Figure 5. Overfitted, appropriately fitted, and underfitted models. A good model should generalize beyond the training points.

Figure 5 gives the intuition behind overfitting and underfitting. An overfitted model follows the training data too closely and can fail on new data. An underfitted model is too simple to capture useful structure. The goal is not maximum complexity, but the best generalization to unseen mechanical engineering cases [4].

Notebook-ready starter workflow: 1) Define the engineering question and the unit of prediction. 2) Build one clean datapoint definition. 3) Split the data by site, asset, or time. 4) Train a simple baseline first. 5) Choose a metric that matches the cost of errors. 6) Inspect the worst predictions with engineering judgment. 7) Only then try more advanced models.

Machine Learning Mathematics

Machine learning can be thought of as building a function that turns inputs into useful outputs. In symbols, we often write this as $\hat{y} = f_{\theta}(x)$ where x is the input, $\hat{y}$ is the predicted output, and θ stands for the model parameters that must be learned from data. During training, the algorithm adjusts those parameters to minimize an objective function, which measures how bad the current predictions are. Lower is better.

For beginners, the safest mental model is this: machine learning is a structured way to learn patterns from examples. In mechanical engineering, those examples might be mechanical component sensor readings, material sample properties, machined surface images, machine-cycle counts, or coolant level histories.

Model: $\hat{y} = f_{\theta}(x)$

Goal of training: choose parameters θ so predictions are useful.

Example objective: minimize average error, such as

$$\text{MSE}(y, \hat{y}) = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Constraint examples

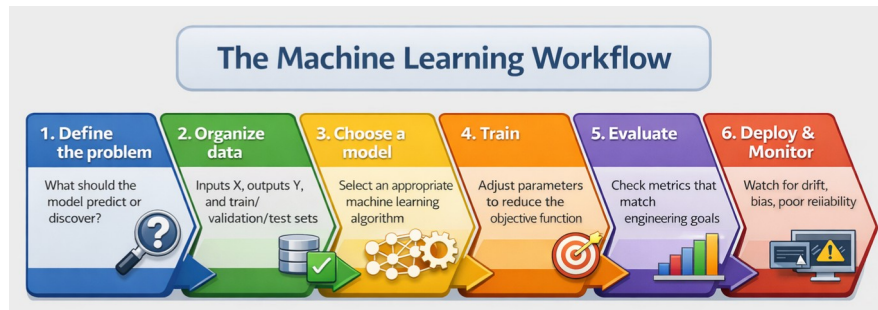
Constraint examples

- Keep predictions physical
- Keep model simple enough
- Respect available data

In a Machine learning workflow, first, you define the problem by deciding exactly what you want the model to predict, such as material strength, production flow, component condition, or bearing stability. Then you organize

the data by collecting the input variables and the correct output values, cleaning errors, and splitting the data into training, validation, and test sets so the model can learn and then be checked fairly.

Next, you choose a model and train it. Choosing a model means selecting a suitable method, such as regression, decision trees, or neural networks, depending on the problem and the data (we will go over such models in the next Modules). Training means the model learns patterns from past examples by adjusting itself to reduce prediction errors. For mechanical engineering students, this is similar to calibrating an analysis model so that it matches measured behavior more closely.



Finally, you evaluate the model and then deploy and monitor it. Evaluation checks whether the model is accurate, reliable, and useful for engineering decisions, especially on new data it has not seen before. Deployment means using the model in practice, but monitoring is still necessary because conditions can change over time, data quality can worsen, and the model may become less reliable. Machine learning is not just about running an algorithm, but about following a careful engineering process from problem definition to long-term use.

Key takeaways

- Machine learning tasks in mechanical engineering often fall into prediction, recognition, feature extraction, or acceleration.
- A datapoint is one consistent unit of observation. Its structure changes across tabular, image, and time-series data.
- Training, validation, and test sets serve different roles. Keep the test set for the final check.
- Regression predicts numbers, classification predicts categories, unsupervised learning finds structure without labels, and semi-supervised learning uses both labeled and unlabeled data.
- Choose metrics by engineering consequence. Accuracy alone is rarely enough.
- Watch for sampling bias, distribution shift, leakage, inconsistent preprocessing, and overfitting.

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